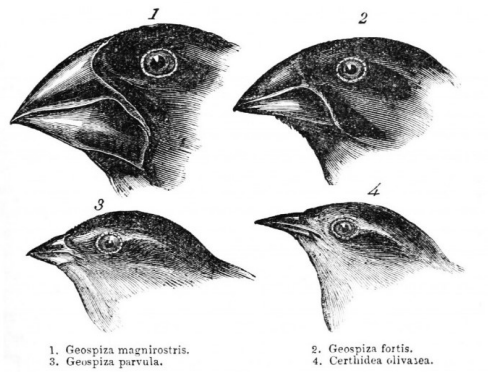


Over long periods of time these changes could be significantly different from what you started with. Yet, no one has ever *witnessed* the origin of a major new animal or plant group

- takes 10,000's or millions of years. We do, however, have an increasing amount of fossil data that shows the evolution of one species from another, step by step and today with molecular techniques.
- We can actually observe and measure the rate of evolution in many species.

Natural selection just selects among whatever variations exist in the population. The result is evolution.

The Theory of Evolution, defined:

"All living species are descendants of ancestral species and are **different** from present day ones due to the cumulative change in the **genetic composition** of a population"

- Sooo in a nutshell, **populations** of living things look and behave differently because over time, their DNA has changed… *but how?*

History of the Theory

- **Evolutionary theory** was developed through many generations of scientists interpreting **new evidence** to refine and expand our understanding of biological change across time.
  - Darwin and Wallace (Evolution)
  - Gregor Mendel (Genetics)
  - Franklin, Watson & Crick (Genetics)

The Nuts and Bolts of Evolution

- **Evolution**: A cumulative change in the inherited characteristics of population.
- **Population** is what changes.
- **Evolution is like a tree** – many branches emerged from a common beginning, some branches died off (extinction), others branched multiple times (present-day diversity)

The Nuts and Bolts of Evolution

- The great diversity of living organisms is the result of over 3.5 billion years of evolution, filling every available **niche** with life forms.
- **Niche**: The area within a habitat occupied by an organism <u>OR</u> the ecological role of an organism within its community.
  - *"I've found my niche in society, I am a social worker"*
  - *"The arctic fish have found their niche in cold waters due to the adaptations in their cell membranes"*

Niche: The area within a habitat occupied by an organism OR the function of an organism within its community.

Discuss: a possible **niche** for

- Elephant
- Whale
- Fish

- *Why* have these life forms filled these niches?
- *How* have these life forms filled these niches?

MICROEVOLUTION

small and often subtle shifts in the genetic makeup of a population from one generation to the next

Microevolution explains why human hair color ranges from blond to black, and why your usual mosquito repellent may suddenly seem less effective one summer.

Convergent evolution occurs when completely unrelated species evolve functionally similar features known as "analogous structures."

↓

understand whether a similar structure found within two different species is analogous is to ask yourself whether their most recent common ancestor also possessed the structure.

Evolution of Whales

It should come as no surprise to learn that sharks and dolphins are very different. Dolphins are mammals, and sharks are fish. A dolphin's skeleton is made of bone, and a shark's skeleton is composed of just cartilage. While dolphins must come to the surface to breathe air, sharks use gills to extract oxygen from the water.

CONVERGENT EVOLUTION

However, both sharks and dolphins have evolved specific analogous traits — streamlined bodies, dorsal fins, pectoral fins and flippers — to achieve the same goal, which is to swim quickly through the ocean and catch prey.

Multiple species from different corners of the world all drawing similar evolutionary conclusions to the challenges and opportunities that they face.
Whether it's soaring through the sky, speeding through the water or trapping prey in sticky pits of doom, instances of convergent evolution are found all throughout nature on many different scales ... and not just in animals, but plants, as well!

The gliding capabilities of the flying lemurs, flying squirrels and sugar gliders

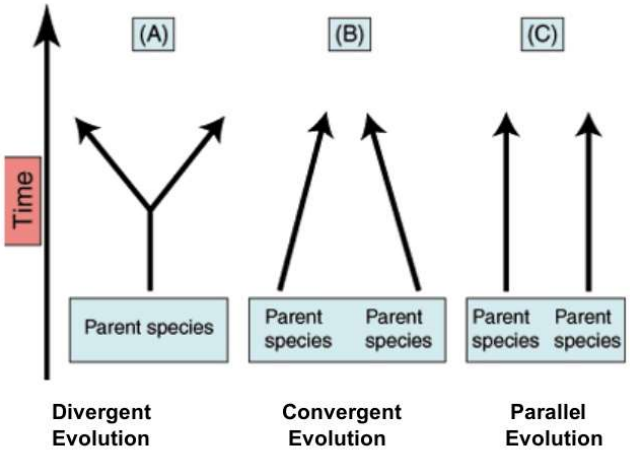
The worm-like bodies of snakes and legless lizards

The pitfall traps of unrelated carnivorous pitcher plant families Nepenthaceae and Sarraceniaceae

The prehensile tails of marsupial opossums and New World monkeys

The ball-shaped bodies of succulents belonging to the unrelated Euphorbia and Astrophytum families

The prickly protusions of echidnas and hedgehogs

DIVERGENT EVOLUTION

Divergent evolution is when two different species share the same ancestral origins but have evolved differently